

Weather Prediction Using Machine Learning

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1. Problem Statement

Accurate weather forecasting is critical for agriculture, disaster management, and transportation. Traditional forecasting relies on complex physical models requiring extensive computational resources. This project explores whether Machine Learning classification models trained on historical weather data can accurately predict whether it will rain the next day (RainTomorrow: Yes/No), making forecasting more accessible and data-driven. I have prior experience building a Smart Weather Monitoring Station using Arduino and C++, which motivated extending this work into the ML domain.

2. Dataset Overview

Property	Details
Source	Kaggle — Rain in Australia
File	weatherAUS.csv
Rows	145,460
Columns	23 (16 numerical, 7 categorical)
Target Variable	RainTomorrow (Binary: Yes/No)
Date Range	2007 – 2017 (10 years)
Missing Values	Present in 20/23 columns
Class Distribution	No Rain: ~78%, Rain: ~22%

3. Models to be Used (In upcoming Phases)

Model	Role
Logistic Regression	Baseline classifier
Decision Tree Classifier	Non-linear patterns
Random Forest Classifier	Primary model

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix

4. Preprocessing Steps Performed

Step	Description
Step 1	Dataset loading and shape verification (145,460 rows x 23 cols)
Step 2	Basic exploration: info(), describe(), head(), tail(), nunique()
Step 3	Duplicate check — 0 duplicates found
Step 4	Null value detection — missing in 20/23 columns
Step 5	Null handling — mode for categorical, median for numerical
Step 6	Label Encoding — 7 categorical columns converted to numeric
Step 7	MinMax Normalization — all numeric features scaled to [0,1]
Step 8	Correlation matrix — feature relationships visualized
Step 9	Dropped Date column — not relevant for ML training
Step 10	Outlier removal — IQR method on Rainfall column
Step 11	Class imbalance check — No Rain: 95,420 vs Rain: 18,228
Step 12	Feature/Target split — X (21 features) and y (RainTomorrow)
Step 13	SMOTE-Tomek — balanced to 95,257 samples per class

5. Libraries & Tools

Category	Tools Used
Language	Python 3
Data Manipulation	Pandas, NumPy
ML & Preprocessing	Scikit-learn
Class Imbalance	Imbalanced-learn (SMOTE-Tomek)
Visualization	Matplotlib, Seaborn
Environment	Jupyter Notebook, VS Code
Version Control	Git & GitHub

6. Challenges Encountered

High Missing Values: Sunshine (48%) and Cloud columns (38-41%) had severe gaps — handled using median/mode imputation to preserve dataset size.

Severe Class Imbalance: ~78% No Rain vs ~22% Rain — resolved using SMOTE-Tomek which oversampled minority class and cleaned borderline samples simultaneously.

Mixed Data Types: 7 categorical columns required Label Encoding before any numerical operations or model training could be performed.

7. GitHub Repository

github.com/muhammad-waleed-dar/Weather-Prediction-ML